

INSTRUCTOR KEY: Grasshoppers and Climate Change

PART 1. How has climate changed along an elevational gradient in the Rocky Mountains of Northern Colorado?

KEY FORMULAS FOR EXCEL SHEET: Yearly Averages & Daily Min & Max
Chaut-C1

Page: March-August Averages

Determining	Column/ Row	Formula
Average	C29	=average(C19:C28)
Average	D29	=average(D19:D28)
Average	E29	=average(E19:E28)
Average	F29	=average(F19:F28)
Use similar logic for D29, E29, F29		
Difference	C31	=C29-C30
Use similar logic for D31, E31, F31		

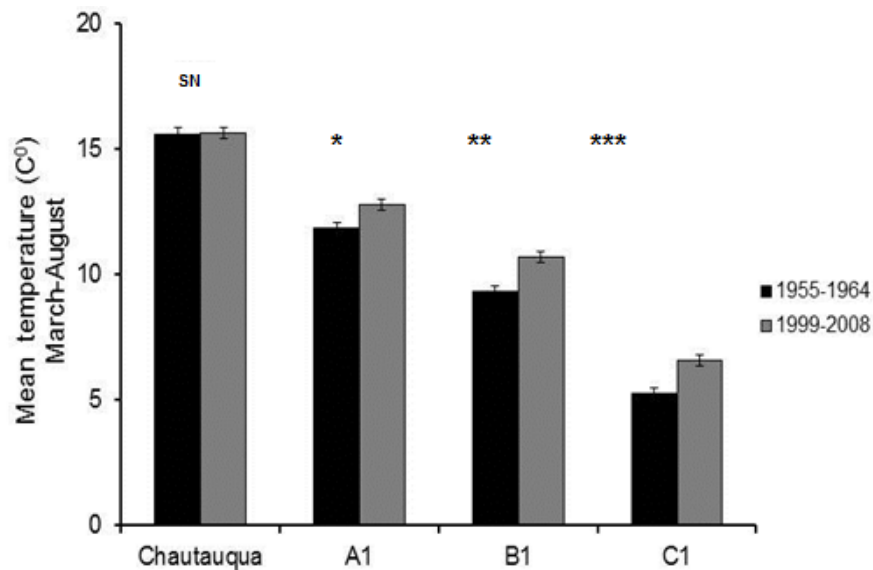


Figure 2. Student groups will draw the 1999-2008 bars next to the ones above, noting difference between the two sampling periods. Differences for Chautauqua, A1, B1 and C1 should be 0.33, 0.96, 1.40, 1.33 C. The average temp at Chautauqua has not changed much but other sites are significantly warmer (see slides). Note that a 1 degree increase means that, on average, each day now is 1 degree warmer than it was previously.

NS = Not Significant, * P = 0.01, ** P = 0.001, *** P = 0.0002.

Part 2. How have grasshopper communities responded to non-uniform warming along the elevational gradient?

Table 1. Grasshopper communities and phenological advancement.

Station	Species	Earliest ordinal date of adult appearance (1959-1960)	Change in timing to adulthood	Number of species advancing
Chautauqua Mesa (1752 m)			-2007-	
	<i>Aeropedullus clavatus</i>	152	-11	
	<i>Melanoplus confusus</i>	155	0	
	<i>Melanoplus sanguinipes</i>	176	7	
	<i>Melanoplus bivittatus</i>	181	2	
	<i>Melanoplus dawsoni</i>	186	-3	
	<i>Hesperotettix viridis</i>	186	4	1/ 6
	Average advancement=		-0.2 d	
Station A1 (2195 m)			-2007-	
	<i>Aeropedullus clavatus</i>	167	-10	
	<i>Melanoplus confusus</i>	167	9	
	<i>Melanoplus dodgei</i>	174	-17	
	<i>Melanoplus sanguinipes</i>	183	7	
	<i>Cratypedes neglectus</i>	195	-26	
	<i>Camnula pellucida</i>	195	10	
	<i>Hesperotettix viridis</i>	202	-12	
	<i>Melanoplus bivittatus</i>	202	3	4/ 8
	Average advancement=		-5.6 d	
Station B1 (2591m)			-2006-	
	<i>Aeropedullus clavatus</i>	172	-13	
	<i>Melanoplus dodgei</i>	172	-13	
	<i>Camnula pellucida</i>	202	-14	
	<i>Circotettix rabula</i>	207	-19	
	<i>Melanoplus dawsoni</i>	215	-27	
	<i>Melanoplus packardii</i>	216	-36	
	<i>Chloealtis abdominalis</i>	216	-22	7/ 7
	Average advancement=		20.6 d	
Station C1 (3048 m)			-2006-	
	<i>Melanoplus dodgei</i>	182	-10	
	<i>Melanoplus fasciatus</i>	202	-15	
	<i>Camnula pellucida</i>	209	-22	
	<i>Chloealtis abdominalis</i>	216	-16	4/ 4
	Average advancement=		-15.8 d	

Negative (-) numbers mean that species have advanced their timing to adulthood, Positive (+) if it is delayed. Must be over 7days to be delayed or advanced, given sampling was done on a weekly basis (5-7 days).

Data Sheet for Chautauqua Mesa, Plains (1,752m or 5,750ft)

Key for 2007 reflects the number of grasshoppers that could be placed within each sampling bin during each sampling period.

Bin number					1	2	3	4	5
Calendar date 2007	13-May	21-May	30-May	4-Jun	13-Jun	18-Jun	25-Jun	2-Jul	9-Jul
Ordinal date	135	141	150	155	164	169	176	183	190

Species in study 2007

<i>Aeropedullus clavatus</i>	No adults found	6	10	14	15	15	20	20	16	116
<i>Melanoplus confusus</i>				6	11	12	10	8	4	51
<i>Melanoplus sanguinipes</i>								8	15	23
<i>Melanoplus bivittatus</i>								12	20	32
<i>Melanoplus dawsoni</i>									20	20
<i>Hesperotettix viridis</i>								12	13	25

Additional species not in the study

1. <i>Arphia conspersa</i>				6	2				8
2. <i>Chortophaga viridifasciata</i>				6	6	4			16
3. <i>Pseudopomala brachyptera</i>					5	4			9
4. <i>Trachyrhynchus kiowa</i>						6			6

Species in study	Ordinal date to adulthood 2007	Ordinal date to adulthood 1959-1960	between surveys (current)
<i>Aeropedullus clavatus</i>	141	152	-11
<i>Melanoplus confusus</i>	155	155	0
<i>Melanoplus sanguinipes</i>	183	176	7
<i>Melanoplus bivittatus</i>	183	181	2
<i>Hesperotettix viridis</i>	183	186	-3
<i>Melanoplus dawsoni</i>	190	186	4

NUMBERS IN GRAY DENOTE NUMBERS PROVIDED ON STUDENT DATA SHEETS. THESE DO NOT HAVE TO BE ADDED TO COLLECTING BINS!

Average of the difference in days between surveys = -0.2
 Number of species advancing (greater than 7 days) = 1 of 6 species

* NOTE: In order to be of consequence, the difference between 1959-1960 & 2007 ordinal dates must be greater than seven because sampling was only done weekly.

Data Sheet for A1, Foothills (2195m or 7201ft)

Key for 2007 reflects the number of grasshoppers that could be placed within each sampling bin during each sampling period.

Bin number					1	2	3	4	5
Calendar date 2007	1-Jun	6-Jun	11-Jun	18-Jun	25-Jun	2-Jul	9-Jul	16-Jul	24-Jul
Ordinal date	152	157	162	169	176	183	190	197	205

Species in study 2007

<i>Aeropedullus clavatus</i>	No adults found	2	3	8	10	20	15	10	8	76
<i>Melanoplus confusus</i>					4	5	6	4	4	23
<i>Melanoplus dodgei</i>		3	5	14	18	14	10	8	5	77
<i>Melanoplus sanguinipes</i>							6	6	8	20
<i>Cratypedes neglectus</i>				5	4	6	8	9	5	37
<i>Camnula pellucida</i>									9	9
<i>Hesperotettix viridis</i>							5	8	12	25
<i>Melanoplus bivittatus</i>									8	8

List additional species not in the study

<i>Xanthipus corallipes</i>					6				6
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Species in study	Ordinal date to adulthood 2007	Ordinal date of adulthood 1959-1960	Difference in days between 1960 and 2007 *
<i>Aeropedullus clavatus</i>	157	167	-10
<i>Melanoplus confusus</i>	176	167	9
<i>Melanoplus dodgei</i>	157	174	-17
<i>Melanoplus sanguinipes</i>	190	183	7
<i>Cratypedes neglectus</i>	169	195	-26
<i>Camnula pellucida</i>	205	195	10
<i>Hesperotettix viridis</i>	190	202	-12
<i>Melanoplus bivittatus</i>	205	202	3

Average of the difference in days between surveys=

-5.6

Number of species advancing (greater than 7 days) =

4 of 8 species

NUMBERS IN GRAY DENOTE NUMBERS PROVIDED ON STUDENT DATA SHEETS. THESE DO NOT HAVE TO BE ADDED TO COLLECTING BINS!

* NOTE: In order to be of consequence, the difference between 1959-1960 & 2007 ordinal dates must be greater than seven because sampling was only done weekly.

Data Sheet for B1, Montane (2591m or 8500ft)

Key for 2007 reflects the number of grasshoppers that could be placed within each sampling bin during each sampling period.

Bin number	1				2		3	4
Calendar date 2007	2-Jun	8-Jun	15-Jun	22-Jun	29-Jun	7-Jul	13-Jul	
Ordinal date	153	159	166	173	180	188	194	

Species in study 2007

<i>Aeropedellus clavatus</i>	No Adults found	9	20	11	19	16	15	90
<i>Melanoplus dodgei</i>		9	8	16	16	11	8	68
<i>Camnula pellucida</i>						6	18	24
<i>Circotettix rabula</i>						8	8	16
<i>Melanoplus dawsoni</i>						5	23	28
<i>Melanoplus packardii</i>					10	14	24	48
<i>Chloealtis abdominalis</i>							6	6

List additional species not in the study

<i>Xanthipus corallipes</i>				8	8			16
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Species in study	Ordinal date to adulthood 2006	Ordinal date of adulthood 1959-1960	Difference between surveys (current survey-previous)
<i>Aeropedullus clavatus</i>	159	172	-13
<i>Melanoplus dodgei</i>	159	172	-13
<i>Camnula pellucida</i>	188	202	-14
<i>Circotettix rabula</i>	188	207	-19
<i>Melanoplus dawsoni</i>	188	215	-27
<i>Melanoplus packardii</i>	180	216	-36
<i>Chloealtis abdominalis</i>	194	216	-22

NUMBERS IN GRAY DENOTE NUMBERS PROVIDED ON STUDENT DATA SHEETS. THESE DO NOT HAVE TO BE ADDED TO COLLECTING BINS!

Average of the difference in days between surveys = -20.6
 Number of species advancing (greater than 7 days) = 7 of 7 species

* NOTE: In order to be of consequence, the difference between 1960 & 2007 ordinal dates must be greater than seven because sampling was only done weekly.

Data Sheet for C1, Subalpine (3048m or 10,000ft)

Key for 2006 reflects the number of grasshoppers that could be placed within each sampling bin during each sampling period.

Bin number		1	2	3	4	5	
Calendar date 2006	14-Jun	21-Jun	29-Jun	6-Jul	12-Jul	19-Jul	26-Jul
Ordinal date	165	172	180	187	193	200	207

Species in study 2006

<i>Melanoplus dodgei</i>	No adults found	10	15	13	14	12	4	68
<i>Melanoplus fasciatus</i>				4	10	6	6	26
<i>Camnula pellucida</i>				7	15	14	15	51
<i>Chloealtis abdominalis</i>						8	6	14

Additional species not in the study

1. <i>Circotettix rabula</i>		3		2	3		8
2. <i>Trachyrachus kiowa</i>				2	1		3
3. <i>Arphia conspersa</i>	8	4	2	4	2		20

Species in study	Ordinal date to adulthood 2006	Ordinal date to adulthood 1959-1960	Difference between surveys (current survey- previous)
<i>Melanoplus dodgei</i>	172	182	-10
<i>Melanoplus fasciatus</i>	187	202	-15
<i>Camnula pellucida</i>	187	209	-22
<i>Chloealtis abdominalis</i>	200	216	-16

NUMBERS IN GRAY DENOTE NUMBERS PROVIDED ON STUDENT DATA SHEETS. THESE DO NOT HAVE TO BE ADDED TO COLLECTING BINS!

Average of the difference in days between surveys =	-15.8
Number of species advancing (greater than 7 days) =	4 of 4 species

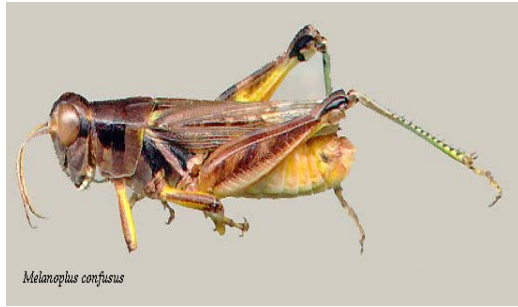
* NOTE: In order to be of consequence, the difference between 1959-1960 & 2007 ordinal dates must be greater than seven because sampling was only done weekly.

Chautauqua Mesa ---

Grasshoppers in survey



Aeropedellus clavatus



Melanoplus confusus



Melanoplus sanguinipes



Melanoplus dawsoni



Melanoplus bivittatus



Hesperotettix viridis

Grasshoppers also encountered but not in survey



Arphia conspersa



Chortophaga virdifasciata



Pseudopomala brachyptera



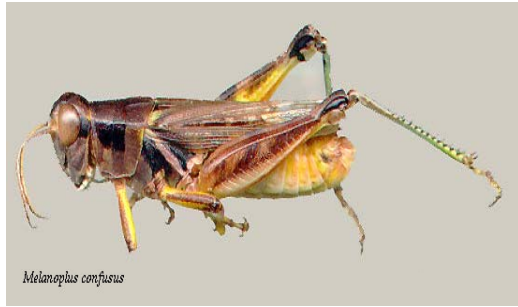
Trachyrhynchus kiowa

Weather station A1 ---

Grasshoppers in survey



Aeropedellus clavatus



Melanoplus confusus



Melanoplus sanguinipes



Melanoplus dodgei



Melanoplus bivittatus



Hesperotettix viridis

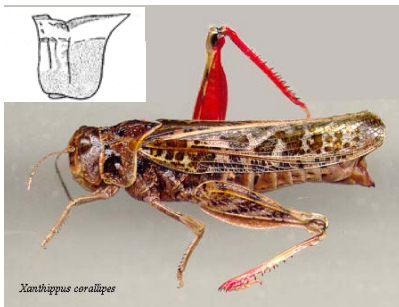


Camnula pellucida



Cratyleptes neglectus

Grasshoppers also encountered but not in survey



Xanthippus corralipes

Weather station B1 ---

Grasshoppers in survey



Aeropedellus clavatus



Melanoplus packardii



Melanoplus dodgei



Chloeaaltis abdominalis



Camnula pellucida

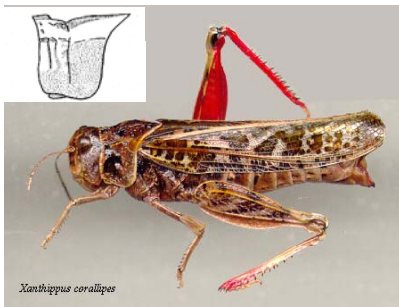


Circotettix rabula



Melanoplus dawsoni

Grasshoppers also encountered but not in survey



Xanthippus corralipes

Weather station C1 ---

Grasshoppers in survey



Melanoplus fasciatus



Melanoplus dodgei



Camnula pellucida



Chloealtis abdominalis

Grasshoppers also encountered but not in survey



Circotettix rabula



Trachyrachus kiowa



Arphia conspersa

PART 3. How might grasshoppers advance their phenology given future estimates of climate changes?

KEY FORMULAS FOR EXCEL SHEET: Yearly Averages & Daily Min & Max
Chaut-C1

Pages: Chautauqua Mesa, A1, B1, C1

Determining	Column/ Row	Formula	Further instructions
Average	Start E5	=average(C5+D5)	Select bottom right of cell & scroll to E369
Daily Growing degree days (GDDs)	Start G5	= E5 -12	Select bottom right of cell & scroll to G369
Daily GDDs, neg values = 0	Start H5	=If(G5<0, 0, G5)	Select bottom right of cell & scroll to H369
Accumulated GDDs	Start K6	=H6+K5	Select bottom right of cell & scroll to K369
For Scenario +1			
Average daily temp + 1	Start M5	=E5+1	Select bottom right of cell & scroll to M369
GDDs	Start N5	=M5-12	Select bottom right of cell & scroll to N369
GDDs, neg values = 0	Start O5	=If(N5<0, 0, N5)	Select bottom right of cell & scroll to O369
Accumulated GDDs	Start Q6	=O6+Q5	Select bottom right of cell & scroll to Q369
For scenario +3 & +5 use same logic as scenario +1 above			

TO PLOT ACCUMULATED GDDs and ORDINAL DATE

In Excel select both column J, rows 4-369 and column K, rows 4-369 and then choose from the top menu “insert → scatter → scatter with smooth lines”. This should produce one curve in **Figure 4**.

TO PLOT DIFFERENT WARMING SCENARIOS (+1, +3, +5) and ORDINAL DATE

In Excel select columns AG rows 4-369 to AK rows 4-369. Choose from the top menu “insert → scatter → scatter with smooth lines”. This should produce all curves as seen in **Figure 5**.

HOMEWORK PAGES: HW Bark Beetles, HW Butterfly

Students should use the logic they learned in the Excel exercises above.

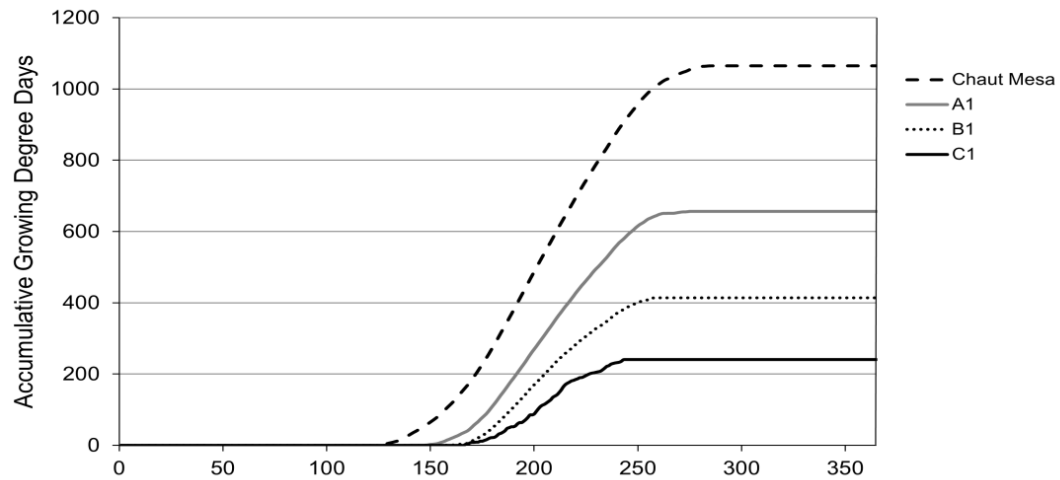


Figure 4. Each group will have a chance to go to the board and draw out their accumulative GDDs pattern. Have the students note the total number of GDDs that occur on average over the last 50 years at each site (Chaut = 1065, A1=656, B1 = 414, C1=241), the ordinal date at which each curve begins (Chaut = ordinal day 124, A1= 139, B1 = 157, C1=167) and when they stop accumulating GDDs (Chaut = ordinal day 285, A1= 275, B1 = 260, C1=243). Have the class describe the pattern and make sure they all draw the curves on the figures.

Notice that GDDs begin accumulating later in the season as one moves up in elevation and that GDD accumulate for a longer time period at lower elevations; making the season longer at lower elevations. Notice also that the number of GDDs that accumulated by the end of the season are higher at lower elevation.

See above for Excel formulae

Figure 5 should look similar to Figure 4, with the curves beginning sooner and ending later as warming increases.

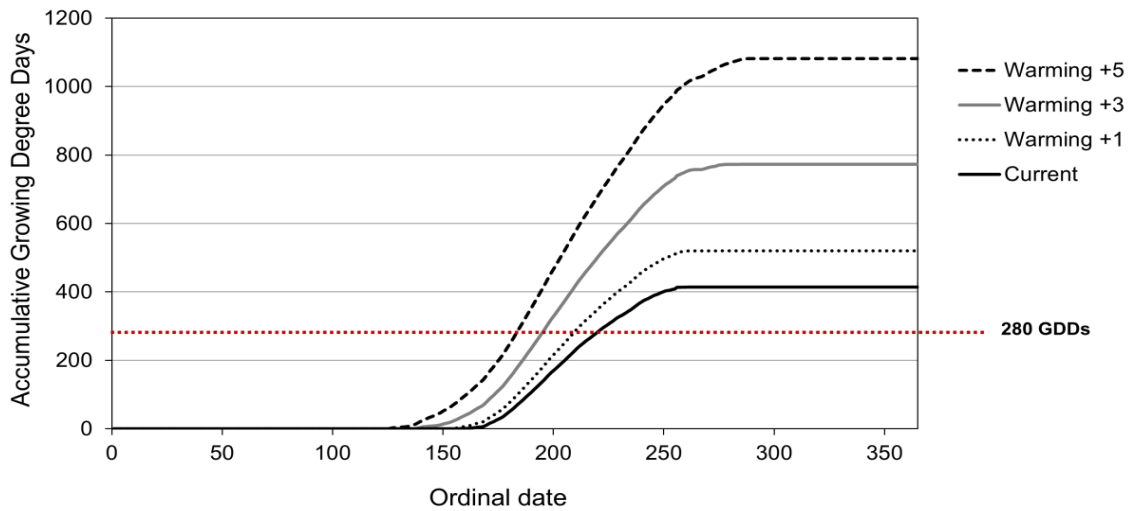


Figure 5. GDD accumulation patterns in the montane life zone (B1, 2591 m) given current climatic conditions and projected +1, +3, and +5°C increases by 2100. The grasshopper of interest requires 280 GDDs at B1, so students can either use the graph to estimate the change in the timing to adulthood or they can look directly at columns AG-AK to determine the exact date associated with 280 GDDs given each warming scenario.